

Denormalized Model (Firestore JSON)

C

CompleteAttendance

key: teacherID_hour

teacher_name: String

classroom: String

schedule: String

status: String

timestamp: DateTime

location: Object

supervisor: Object

Direct access
No JOINS
All information in a single node

Normalized Model (SQL)

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Teacher

name: String

teacherID: String (PK)

1

requires JOIN

0..*

C

ClassSession

classroom: String

schedule: String

classID: String (PK)

teacherID: String (FK)

1

requires JOIN

0..*

C

Attendance

status: String

timestamp: DateTime

attendanceID: String (PK)

classID: String (FK)

Complete query requires:
SELECT * FROM Teacher p
JOIN ClassSession c ON p.teacherID = c.teacherID
JOIN Attendance a ON c.classID = a.classID